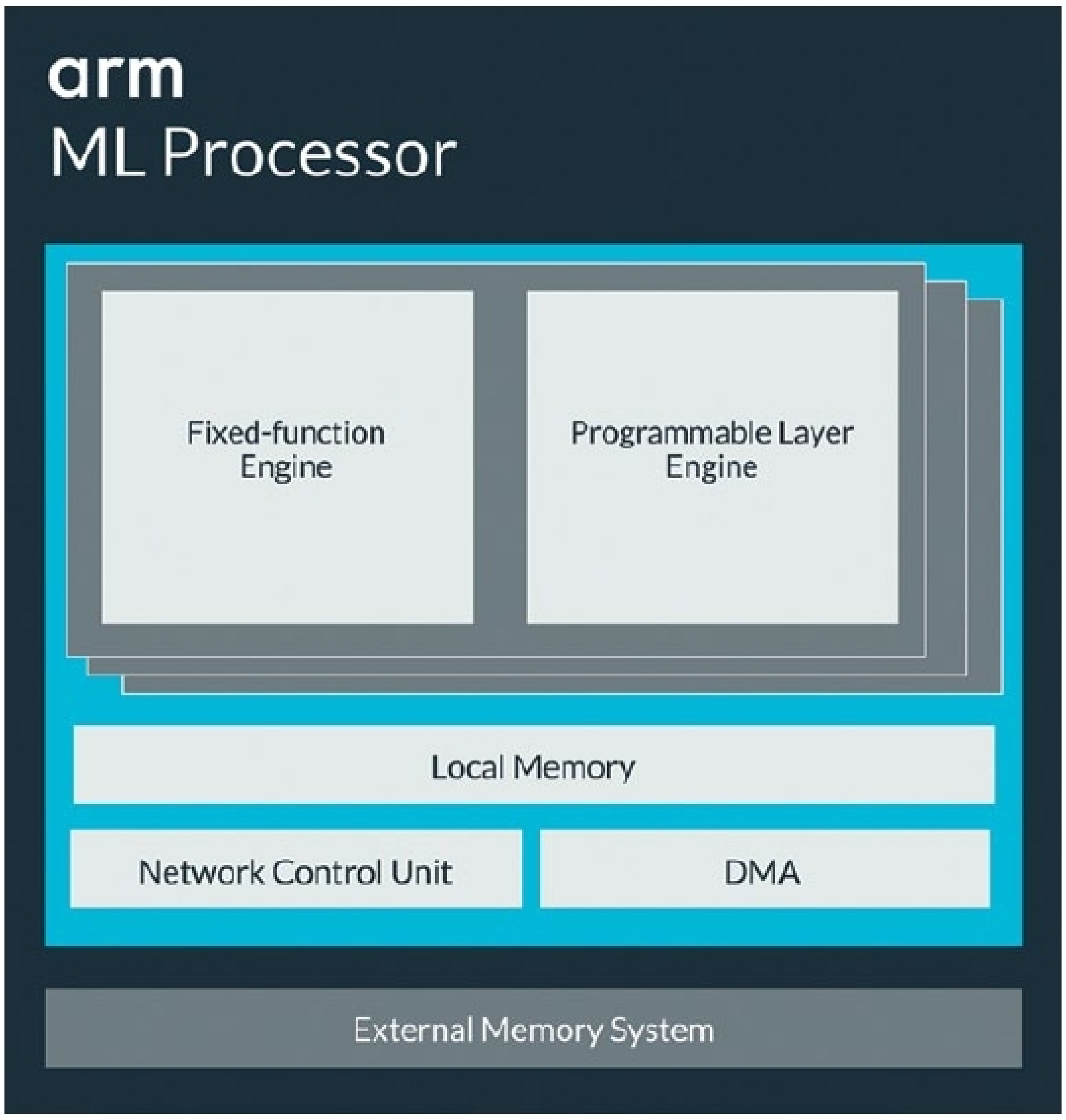
designing methods and approaches easier and more advanced for designers. Neural networks, deep ML, augmented reality (AR), virtual reality (VR) and mixed reality (MR) are some of the outstanding advancements that AI has made in design and development industries.

Why move towards AI-powered devices

AI at the Edge is enabling organisations to use massive amounts of data collected by sensors and devices for smart manufacturing, healthcare, aerospace and defence, transportation, telecom and cities to provide engaging customer experiences. Sensors collect such data as sound, vibration, movement and temperature, and analyse it to notify about relevant occurrences in real time. Rise of autonomy in Edge environments, like smart home assistants and autonomous robots, requires the most advanced computing platforms. Other areas that can use AI's potential in an innovative way are:

- Neural networks can utilise AI to identify various behaviours and use this technology in Edge computing (sensors) to make themselves smart and self-reliant.
- AI can be used in aviation applications to greatly decrease runway inspection time, increase obstacle detection and reduce chances of flight delays.
- AI-based automated optical inspection systems can deliver higher levels of inspection and improve inspection accuracy in manufacturing applications.
- AI at the Edge can be used for traffic monitoring and analytics at road intersections can help avoid road accidents.
- In retail and logistics applications, autonomous mobile robots powered by AI can pick and prepare orders, replenish store shelves and deliver packages to customers.



Arm ML processor (Credit: www.arm.com)

Role of computing processors

AI redefines device capabilities through the growing design and development ecosystem that has extended its reach from smartphone central processing unit (CPU) design and app development to ML. ML engines process metadata from operating systems and applications to detect cybersecurity threats, and optimise operations of devices and networks in which these operate.

AI-powered devices built with CPUs and microcontrollers (MCUs) handle much of AI and ML workloads at the Edge. Their architecture supports diverse AI applications and the specific requirements these need to run. Extremely power-efficient MCUs are required for AI algorithms in highly-constrained, battery-powered Edge devices, such as wearables and sensors. High performance, energy-efficient media processors are required for mobile and consumer devices like tablets, smartphones, cameras and smart TVs.

The Internet of Things (IoT) platforms, combined with a new class of advanced and ultra-efficient ML processors, are transforming the IoT into a global network of securely-managed devices.

An ML processor's optimised design enables new features, improved user experience and delivers innovative applications for a wide range of market segments. It provides a massive uplift in efficiency compared to CPUs, graphics processing units (GPUs) and digital signal processors (DSPs) through efficient convolution, sparsity and compression.

An onboard processor and AI engine can increase computing speed. By integrating an advanced image signal processor into an AI engine, the platform's processing time can be improved, as it does not need to communicate as much with a distant server. Enabling devices to process data locally reduces the burden on networks, which is important as the number of connected devices continues to grow.

"Recent Edge computing innovations equip algorithm-powered devices with complicated computation and storage capabilities to perform mission-critical functions that are closer to the point of impact. An integrated model of cloud and Edge computing working hand in hand allows us to optimise compute, store and network resources, and deliver the best AI performance," says Dr Yangdong Deng, Tsinghua University.

Impact of AI on the design industry

Learning and keeping up with advanced designing tools and methods have now become crucial for designers. Without knowing these advanced tools and software, it is impossible for them to succeed.

Technologically-advanced tools